

# WILL MCGOWAN

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## EDUCATION

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University of Strathclyde, Glasgow

Sep 2023 - Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1
- **Relevant Modules:** Applied Econometrics; Treasury Management and Derivatives; Industrial Economics

**Dissertation** | MATLAB

In progress, due 2027

*"Does the News Pay? Forecasting European Gas Volatility with MIDAS, Storage and Text Data"*

- Testing whether news attention improves forecasts of TTF gas price volatility with mixed-frequency (MIDAS) models built on daily storage, market and news data, including an original Google Trends attention index
- Benchmarking out-of-sample against HAR and AR models with Diebold-Mariano tests across pre-crisis, crisis and post-crisis periods, and measuring economic value through the Sharpe ratio of a volatility-targeted TTF futures strategy

**Empirical CAPM Test**

MATLAB

- Tested CAPM on 704 anomaly portfolios across 56 countries (JKP global factor data): zero alpha rejected for 32% at 5% and jointly (GRS) in 48 of 52 countries, with momentum and value the most persistent

**Monte Carlo Option Pricing Engine**

MATLAB

- Built a vectorised Monte Carlo engine pricing European and Asian options under risk-neutral GBM, benchmarked against Black-Scholes and the closed-form geometric-Asian price
- Reduced the standard error of arithmetic-Asian estimates by 97.24% vs plain Monte Carlo at  $N = 100,000$  using a geometric-Asian control variate, alongside antithetic variates

**Corn-Soybean Cointegration** | group project

R

- Found one cointegrating relationship between corn and soybean prices (quarterly, 1990-2022) with Engle-Granger and Johansen tests; the error-correction model showed both prices adjusting back to equilibrium (soybean coefficient -0.29,  $p < 0.001$ )

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

## WORK EXPERIENCE

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Le Set, Glasgow | Head Mixologist

Sep 2026 - Present

- Created 4 original house cocktails and wrote the specifications for the 7 classics on the menu of a new 200-cover restaurant and cocktail lounge, opened Sep 2026
- Wrote batch specifications for 5 prebatched cocktails and trained around 15 bar staff to prepare and serve them consistently

KONG, Glasgow | Mixologist and Front of House

Oct 2024 - Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%
- Reconciled cash and card takings at the end of each shift and trained new staff on drink specifications, POS workflows and licensing compliance

## LEADERSHIP AND ACTIVITIES

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Strathclyde Agricapital Society | Co-Founder and Treasurer

Sep 2025 - Present

- Co-founded the university's first agri-finance society, grew membership to 126 and led weekly research sessions on deals such as Olam Agri's USD 1.24bn stake sale to SALIC
- Built a wheat calendar-spread strategy in R from futures curve structure and CFTC managed-money positioning; weekly backtest (Apr 2025 - Oct 2026): Sharpe 0.76 before costs and 0.28 after, maximum drawdown 3.1%

Strathclyde Poker Society | Vice-President, then President

Sep 2025 - Apr 2026

- Ran weekly tournaments for around 60 players on a £600-per-semester budget, reconciling every buy-in and prize pool

## ADDITIONAL INFORMATION

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**Technical:** R, MATLAB, Python, Excel, Git

**Interests:** chess, 14th at the Irish U19 Chess Championships (2023) and won 21 of 22 games in a simultaneous exhibition against players averaging 1860 FIDE; powerlifting, Northern Ireland U19 champion (2023)